

Content Orchestrator

Engineering Milestone 3

Independent Audit Report

Prepared: July 25, 2026

EXECUTIVE SUMMARY

Milestone 3 of the Content Orchestrator platform has been independently audited, verified, and cleared for review. The engineering team delivered a fully working orchestration engine – the backbone that will automate the end-to-end production of AI-generated video content from idea through publish. All quality and security gates passed with zero outstanding blockers.

WHAT WAS BUILT

Orchestration Engine

A state machine that drives each piece of content automatically through every production stage (scripting, voiceover, visuals, review, publish). It handles retries, spend limits, and human checkpoints without manual intervention.

Worker Framework

A background worker service that picks up production tasks, executes them, and reports results – designed to plug in real AI providers (OpenAI, ElevenLabs, etc.) in the next milestone.

Database & Security Foundation

24 database migrations applied cleanly. Every table is protected by Row-Level Security – users can only see and modify data belonging to their own workspace. Three new security-bypass helper functions prevent infinite-loop vulnerabilities.

Spend Controls

Per-workspace daily and monthly spend caps are enforced before any AI call is dispatched. Runs are automatically paused when a budget threshold is reached.

Human Review Gate

Any production stage can require human sign-off before the pipeline advances. Rejection routes the run back for correction.

AUDIT RESULTS

Tests:	39 / 39 automated integration tests pass against a real database. Coverage 85 % (gate: 70 %).
Code Quality:	Zero lint violations. All modules compile cleanly.
Security:	No secrets in codebase or version history. All 28 tables enforce mandatory row-level security. JWT validated (audience, subject, expiry).
Migrations:	All 24 migrations apply cleanly from blank database; single linear chain, no conflicts.
CI Pipelines:	All three automated build pipelines (API, Worker, Web) pass on GitHub Actions.

DEFECTS FOUND & FIXED

Self-Leave Permission Bug - High

A database security policy only allowed administrators to remove members. When a non-admin user tried to leave a workspace, the system returned a success response but silently failed to remove them - the membership remained active. Fixed: the policy now explicitly permits a member to delete their own record.

CI Registry URL Bug - High

The dependency lock file contained internal URLs unreachable from GitHub's build servers, causing the web CI job to fail. Fixed: all 260 package URLs rewritten to the public npm registry.

Stale Test Data - Medium

Several tests were picking up leftover database rows from previous runs, causing intermittent failures. Fixed: each test now cleans up relevant rows before running.

DELIVERY METRICS

Repository:	github.com/royalindustry94-crypto/Content-orchestrator
Branch / PR:	feature/milestone-3 - Pull Request #1 (open, not merged)
Final Commit:	7689ae1f
Tests:	39 / 39 passed
Coverage:	85 % (gate: 70 %)
Migrations:	24 (head: 0024, single linear chain)
Files Changed:	88 files, +11,222 / -76 lines vs main
Security Findings:	0 critical, 0 high remaining after fixes
Blockers:	None

NEXT STEPS

Milestone 4 - AI Provider Integration

Connect real AI providers (OpenAI for scripting, ElevenLabs for voiceover, image generation for visuals). The orchestration engine is designed to accept these integrations with minimal changes.

Merge Decision

Pull Request #1 is open and awaiting approval to merge. No technical action is required - this is a sign-off checkpoint only.